

Can a Machine Learn to Recycle?

A DS 4002 Case Study

The Problem

Only about 32% of solid waste in the United States is actually recycled, the rest ends up in landfills, even when much of it could have been diverted. A major reason is incorrect sorting: when non-recyclable items are mixed into recycling bins, they can contaminate entire batches, leaving them unprocessable. Human error is hard to eliminate, but what if a machine could do the sorting automatically? Real-world waste sorting robots already exist in some facilities, and the underlying technology (computer vision powered by deep learning) is what makes them possible. Convolutional Neural Networks (CNNs) have proven remarkably effective at learning visual patterns: edges, textures, shapes, and colors that distinguish one material from another. The question is whether a CNN trained on real landfill images can reliably tell recyclable waste from non-recyclable waste.

Your Role

You are a machine learning engineer contracted by a waste management company piloting an automated sorting system at their facility. They have provided you with the RealWaste dataset, thousands of images of actual waste items photographed at the point of reception at a landfill in Wollongong, Australia. Your job is to build and evaluate a binary image classification model that can look at a photo of a piece of waste and determine: Is this recyclable or not?

Your Mission

Using the RealWaste dataset and Python, you will preprocess the image data, define a binary recyclable/non-recyclable labeling scheme, and train at least two CNN-based models, a custom baseline architecture and a transfer learning approach using a pretrained network. You will evaluate your models, compare their performance, and deliver a report with your findings and recommendations to the waste management company. Full deliverable details and success criteria are outlined in the accompanying rubric.

Data source: RealWaste Dataset, UCI Machine Learning Repository (2023). Images captured at Whyte's Gully Waste and Resource Recovery Centre, New South Wales, Australia.